

Real Applications of Bayesian Statistics

STAT 521 – Applied Bayesian Inference

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Bayesian Methods with Large Data

Even when the prior becomes negligible, Bayesian methods remain powerful because they provide tools that frequentist methods do not:

- **Posterior predictive distributions**
Critical for forecasting and decision-making.
- **Coherent uncertainty quantification**
Essential for complex or hierarchical models.
- **Hierarchical modeling**
Borrowing strength across groups remains valuable at scale.
- **Latent variable, mixture, and topic models**
Bayesian approaches shine in high dimensional latent structure.

Bayesian Statistics Applications

Bayesian methods are increasingly used in:

- Government and public policy
- Medicine and clinical trials
- Technology and large-scale machine learning
- Sports analytics and real-time decision systems
- Finance, risk modeling, and insurance
- Robotics and self-driving vehicles
- Environmental and climate sciences

FDA and Bayesian Adaptive Trials

Bayesian methods are formally accepted by the FDA for:

- Medical device approval
- Early-phase drug trials
- Adaptive trial design
- Dose escalation and early stopping

Why Bayesian?

- Allows borrowing information from previous studies
- Provides real-time updating during a trial
- More ethical: fewer patients exposed to inferior treatments

FDA Guidance on Bayesian Clinical Trials¹

¹<https://www.fda.gov/media/71512/download>

Example: Medtronic Cardiac Ablation Device

Real case: a medical device was approved using a **Bayesian hierarchical model**.

- Pooled prior information from historical trials
- Posterior predictive probability used for approval
- Reduced required sample size

This is a concrete example of Bayesian models influencing regulatory decisions.

Bayesian models are operationally used for:

- Small-area estimation (SAE)
- Poverty and unemployment estimates
- Education indicators
- Health and disease burden estimation

Why Bayesian?

- Stabilizes estimates in areas with very small or no samples
- Hierarchical shrinkage borrows strength across regions and subgroups
- Directly quantifies uncertainty to support policy decisions

Census Bureau SAE Model Specification²

²<https://www.census.gov/topics/research/stat-research/expertise/small-area-est.html>

Example: U.S. County-Level Poverty Rates

Bayesian Fay–Herriot model is used to:

- Estimate poverty rates for all U.S. counties
- Allocate billions of dollars in federal funding
- Provide stable estimates even when sample sizes are tiny

A true example of Bayesian methods impacting social policy and resource allocation.

Weather Predictions

- Weather is inherently uncertain, Bayesian models produce **probabilistic forecasts** (full distributions).
- Bayesian methods combine multiple data sources and forecast models to account for uncertainty and bias.
- Allows for coherent **uncertainty quantification** and risk-aware decision making (e.g., extreme event warnings).
- Offers systematic framework for **model averaging** and calibration of ensemble forecasts.

Modern weather prediction uses ensembles:

- Many runs of the same model with different initial conditions, or
- Multiple different models.

Bayesian model averaging assigns data-driven weights to each model.

Sports Analytics (MLB, NBA, Soccer)

Bayesian models are widely used in modern sports analytics for:

- Player ability estimation using hierarchical priors (e.g., batting averages, shooting percentages)
- Real-time win probability updating as new plays occur

Why Bayesian?

- Naturally supports dynamic updating as games unfold (online inference)
- Shrinks noisy estimates toward league averages, improving accuracy for players with limited data

Google, Amazon, and Meta use Bayesian optimization

Key idea: *Bayesian optimization is not about modeling huge datasets.* It is about efficiently optimizing a black-box function that is **expensive to evaluate**. The Bayesian model governs uncertainty over the objective function.

Bayesian optimization is ideal when:

- The search space is high-dimensional
- Evaluations are expensive (compute, time, or risk)
- Only a small number of trials is possible

Bayesian Optimization: A Different Use of “Bayesian”

Bayesian optimization is used when each function evaluation is:

- **Costly** (e.g., training a model takes hours)
- **Slow** (A/B tests need days of traffic)
- **Risky** (search ranking experiments impact billions of users)

The Bayesian prior is over the **unknown function** we want to optimize.

Because we only get a small number of function evaluations, the prior:

- guides exploration,
- controls uncertainty,
- and remains influential throughout the optimization process.

Bayesian A/B Testing (Google, Meta, Amazon)

Bayesian A/B testing is deployed at scale in search ranking, ads, recommendation systems, and UI experiments.

In frequentist A/B testing, repeatedly checking the p-value inflates the Type I error rate, so early stopping requires complex corrections.

Bayesian methods are widely used for online experiments because they provide:

- **Real-time updating** as data arrive (no fixed sample size)
- **Posterior probabilities** of A beating B (intuitive decisions)
- **Early stopping rules** without inflating Type I error
- **Continuous monitoring** without p-value penalties

Self-Driving Cars: Bayesian Filtering

Bayesian filtering methods (Kalman, Extended Kalman, Unscented Kalman, Particle Filters) are widely used for:

- Localization
- Object tracking
- Sensor fusion (LIDAR + Radar + Camera)

Why Bayesian?

- Real-time uncertainty quantification
- Continual updating as new sensor data arrives
- Robust state estimation for safety-critical navigation

Note: Bayesian filtering is the *dominant* framework for state estimation in autonomous vehicles, though some systems also incorporate deep learning or hybrid methods.